CYBER-SWARM//

HTTPS://CYBERSWARM.VERCEL.APP/

QUANTUM HUB

INTRODUCTION

CYBER-SWARM is an intelligent multi-agent robotic system built to demonstrate how a group of autonomous robots can communicate, collaborate, and make decisions collectively without relying on a single central controller.

The platform combines wireless communication, autonomous task allocation, real-time decision-making, and coordinated navigation to create a scalable robotic swarm.

Each robotic agent can perceive its environment, share relevant information with other agents, evaluate available tasks, and contribute to the swarm's overall objective.

Core Capabilities

- Autonomous Decision Making
- Robot-to-Robot Communication
- Dynamic Task Allocation
- Collaborative Navigation
- Real-Time Swarm Coordination
- Decentralized Control

CYBER-SWARM transforms a collection of individual robots into a coordinated, intelligent system capable of adapting to changing tasks and environments.

"One robot can act. A swarm can think."

Key idea:

Communication → Decision → Coordination → Autonomous Execution

THE CHALLENGE

As the number of autonomous robots increases, coordination becomes more complex than individual robot control. A swarm operating in a real-world environment must continuously make decisions despite changing conditions, limited communication, and competing tasks.

Key Challenges

01 — Communication Constraints

Robots must exchange information reliably while operating as distributed agents. Communication delays or failures can affect collective decisions.

02 — Intelligent Task Allocation

When multiple tasks appear simultaneously, the swarm needs to determine which robot should perform which task based on factors such as distance, availability, and current workload.

03 — Decentralized Coordination

A single central controller can become a bottleneck or point of failure. Individual agents need to participate in decision-making while maintaining a common objective.

04 — Dynamic Environments

Obstacles, robot positions, task priorities, and environmental conditions can change continuously. Static instructions are therefore insufficient.

05 — Collision & Conflict Avoidance

Multiple robots moving simultaneously can create conflicts. The swarm must coordinate movement and adapt routes to avoid collisions and unnecessary interference.

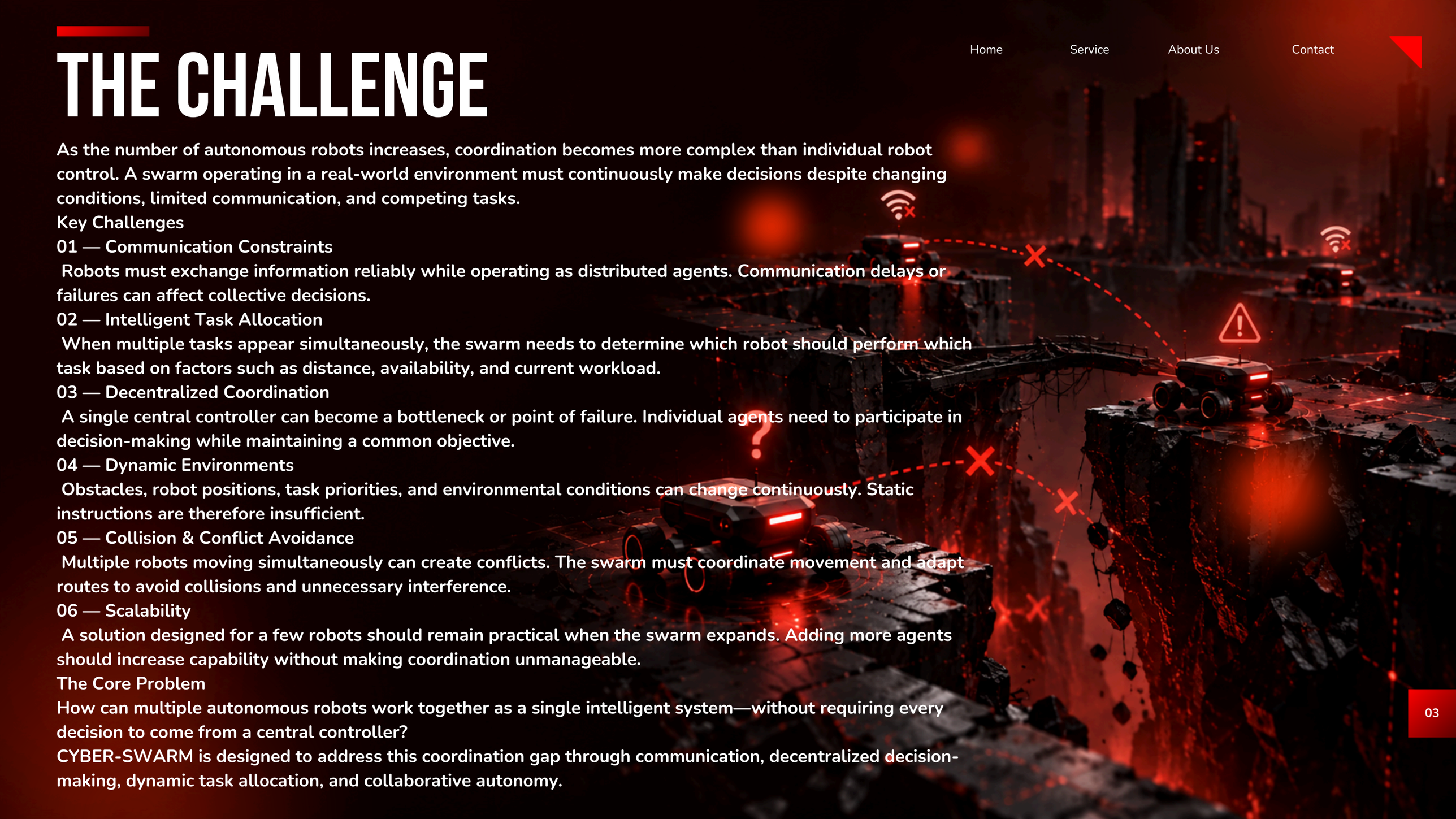
06 — Scalability

A solution designed for a few robots should remain practical when the swarm expands. Adding more agents should increase capability without making coordination unmanageable.

The Core Problem

How can multiple autonomous robots work together as a single intelligent system—without requiring every decision to come from a central controller?

CYBER-SWARM is designed to address this coordination gap through communication, decentralized decision-making, dynamic task allocation, and collaborative autonomy.



THE PLATFORM

CYBER-SWARM is a unified platform that enables multiple autonomous robots to operate as a coordinated swarm rather than as isolated machines.
The platform combines real-time monitoring, wireless communication, intelligent task allocation, and autonomous navigation into a single ecosystem.

PLATFORM ARCHITECTURE

01 Swarm Agents

Each robot acts as an independent agent capable of sensing, communicating, and executing assigned tasks.

02 Communication Layer

Robots exchange information with nearby agents, allowing the swarm to maintain a shared understanding of the environment.

03 Task Allocation Engine

Incoming tasks are evaluated and dynamically assigned to the most suitable available robot.

04 Decision & Coordination Layer

Agents make local decisions while continuously coordinating with the rest of the swarm.

05 Live Monitoring Dashboard

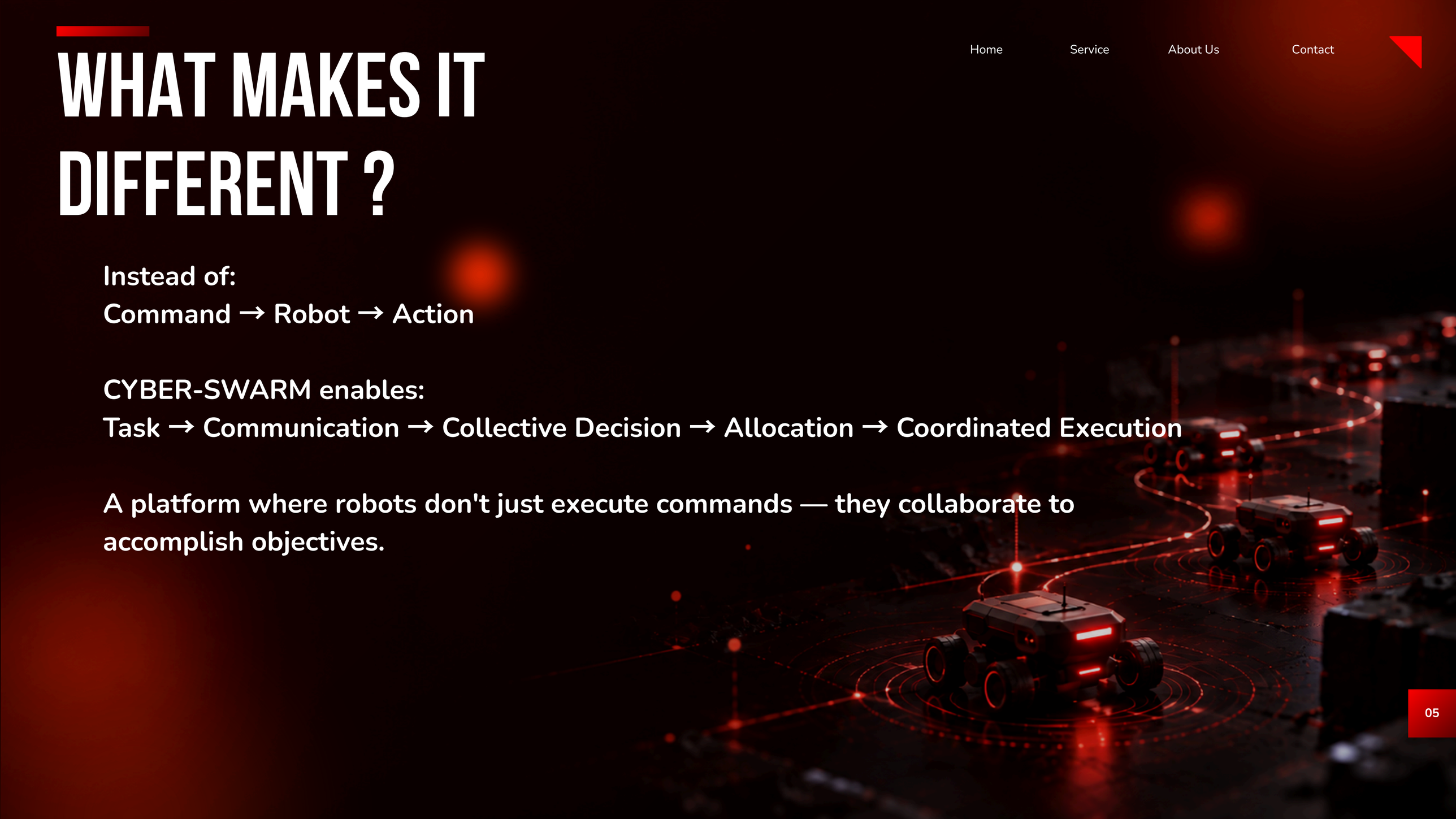
The platform provides real-time visibility into robot status, active tasks, routes, system metrics, and swarm activity.

WHAT MAKES IT DIFFERENT ?

Instead of:
Command → Robot → Action

CYBER-SWARM enables:
Task → Communication → Collective Decision → Allocation → Coordinated Execution

A platform where robots don't just execute commands — they collaborate to accomplish objectives.



PUBLIC EXPLORATION

CYBER-SWARM is not just a robotic system—it is an interactive platform that makes swarm intelligence observable and understandable.
The Public Exploration interface allows users to explore the swarm's behaviour, understand how tasks are distributed, and observe how individual robots contribute to collective objectives.

01 Live Swarm Status

Monitor the current state, availability, and activity of individual robotic agents.

02 Task Allocation Engine

Observe how tasks are created, evaluated, and assigned to suitable robots.

03 Robot Movement

Monitor the current state, availability, and activity of individual robotic agents.

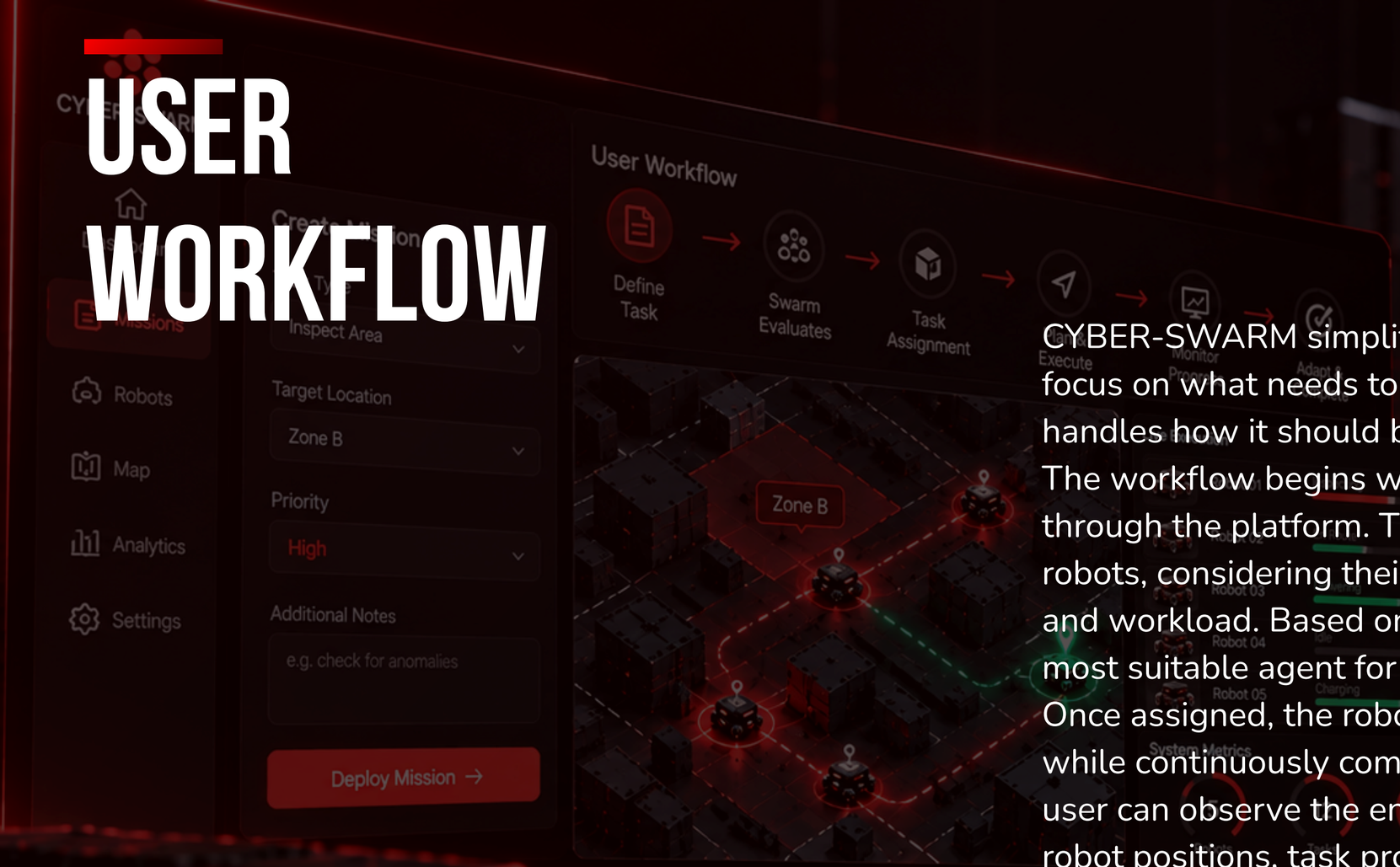
04 Collective Decision-Making

Understand how robots exchange information and coordinate actions instead of operating independently.

05 System Metrics

Explore operational data such as active robots, completed tasks, communication status, and overall swarm performance.

USER WORKFLOW



CYBER-SWARM simplifies swarm operation by allowing the user to focus on what needs to be achieved, while the robotic system handles how it should be executed.

The workflow begins when the user defines an objective or task through the platform. The swarm then evaluates the available robots, considering their current position, availability, capabilities, and workload. Based on this information, the system determines the most suitable agent for the task and assigns it dynamically.

Once assigned, the robot plans its movement and begins execution while continuously communicating with the rest of the swarm. The user can observe the entire process through the platform, including robot positions, task progress, routes, and system activity.

If the environment or task conditions change, the swarm can adapt its decisions and modify its execution accordingly, allowing the operation to continue without constant manual intervention.

DEFINE → DECIDE → ASSIGN → EXECUTE → MONITOR → ADAPT

The user defines the objective. The swarm coordinates the execution.

DATA QUALITY

CYBER-SWARM relies on accurate, consistent and real-time data to support reliable swarm decisions.

Collect
Telemetry, sensors, tasks, system events

→

Process
Filter, validate and remove noise

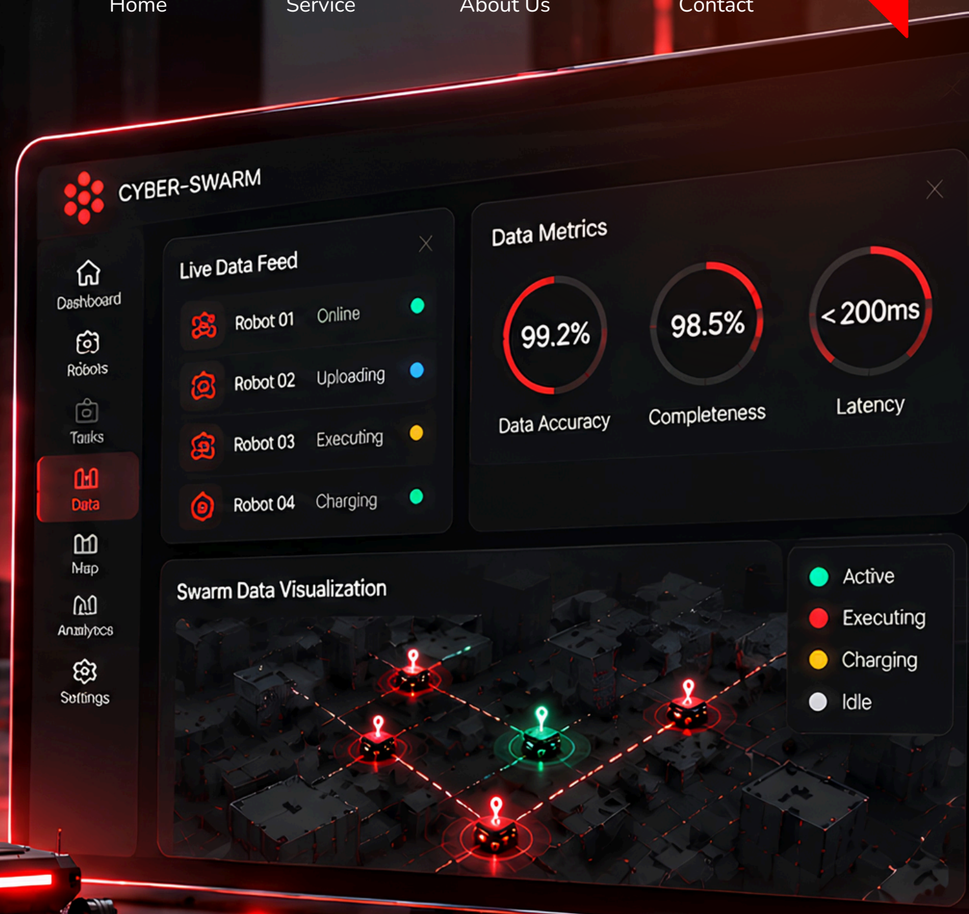
→

Store
Structured and scalable data

→

Visualize
Live insights on platform

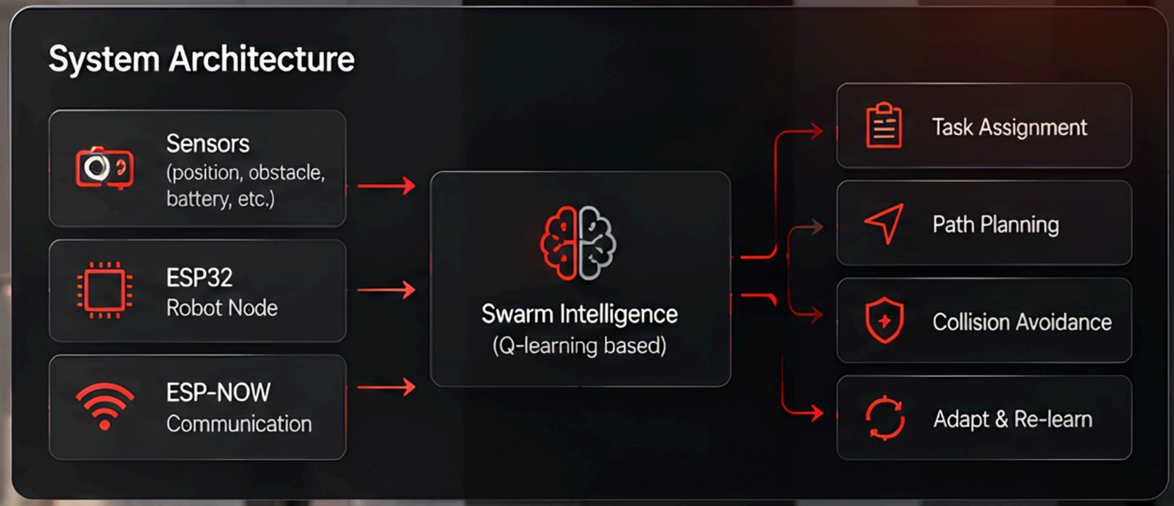
Clean data → Better decisions → Reliable coordination → Smarter swarm



IMPLEMENTATION

A decentralized and scalable system built using ESP32, ESP-NOW and reinforcement learning.

System Architecture



ESP32

- On-board control
- Sensor integration
- Low power & cost-effective

ESP-NOW

- Peer-to-peer communication
- Low latency
- No external network required

AI / Q-learning

- Learns optimal actions
- Dynamic task allocation
- Adapts to environment changes

Sense → Learn → Coordinate → Execute → Improve

CURRENT STATUS

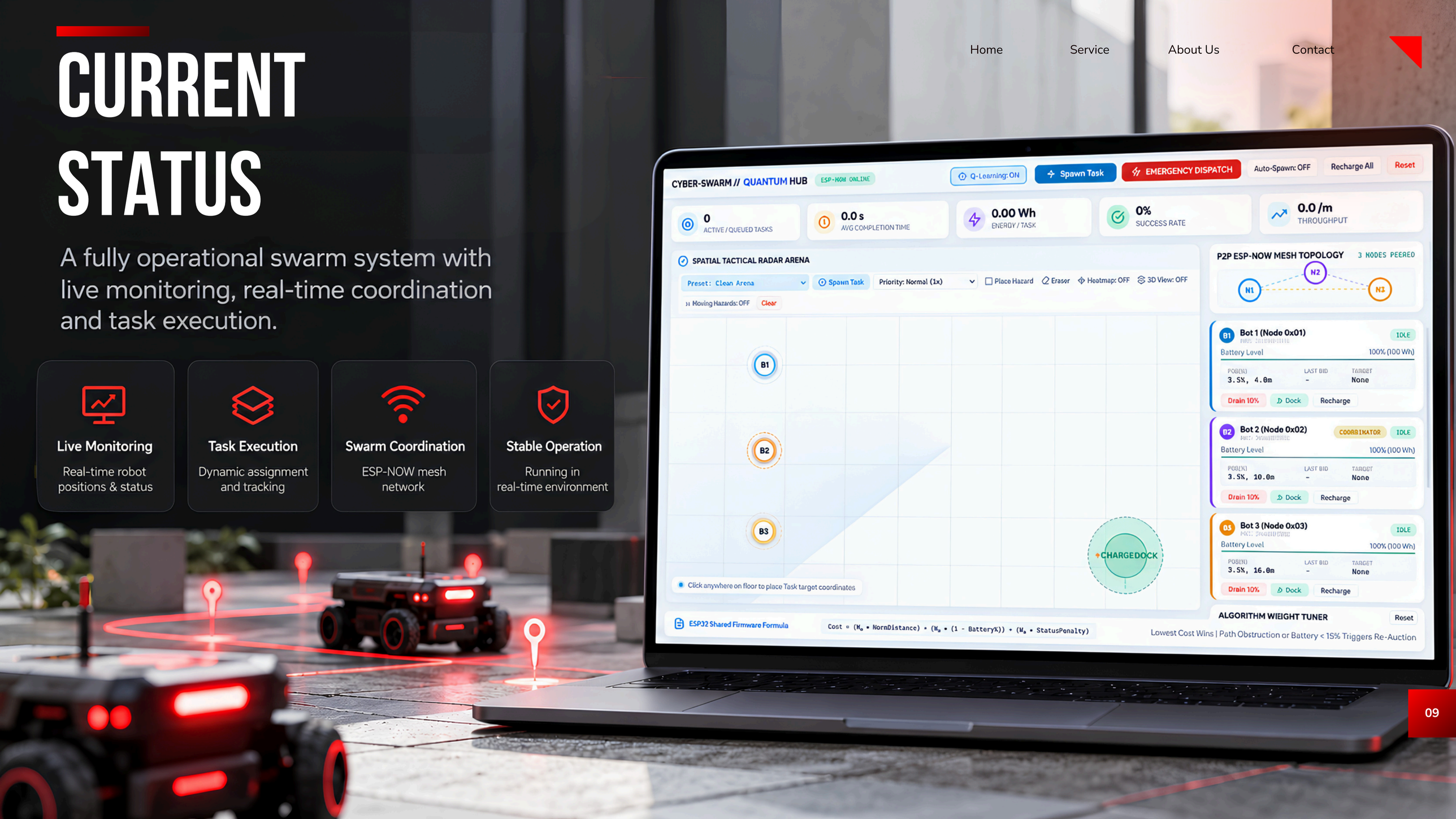
A fully operational swarm system with live monitoring, real-time coordination and task execution.

Live Monitoring
Real-time robot positions & status

Task Execution
Dynamic assignment and tracking

Swarm Coordination
ESP-NOW mesh network

Stable Operation
Running in real-time environment



THANK YOU

CYBER-SWARM

From individual robots to collective intelligence.
An autonomous swarm platform built around real-time communication, intelligent task allocation, decentralized decision-making, and adaptive coordination.

Sense → Communicate → Decide → Coordinate → Execute

Thank you for exploring CYBER-SWARM.

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